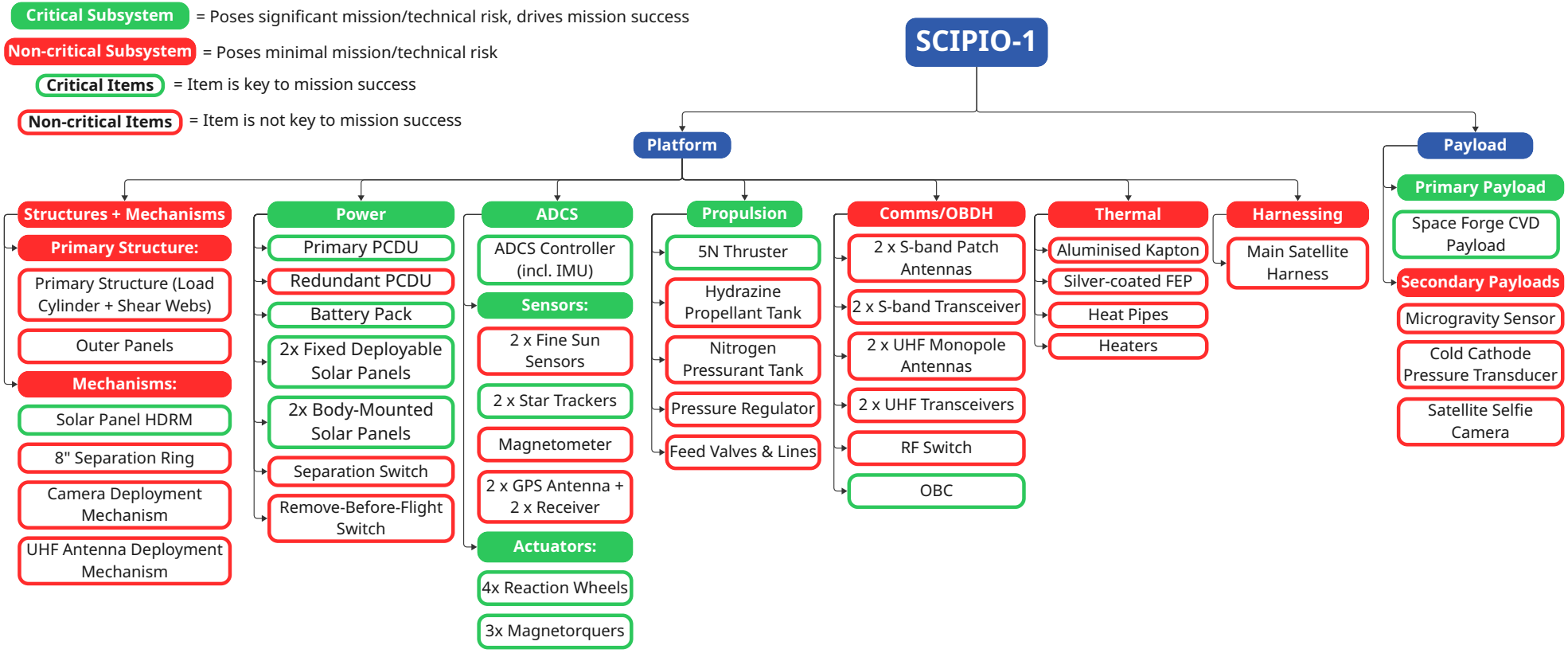

Critical Subsystem = Poses significant mission/technical risk, drives mission success
Non-critical Subsystem = Poses minimal mission/technical risk
Critical Items = Item is key to mission success
Non-critical Items = Item is not key to mission success
SCIPIO-1
Platform
Payload
Structures + Mechanisms
Primary Structure:
Primary Structure (Load Cylinder + Shear Webs)
Outer Panels
Mechanisms:
Solar Panel HDRM
8" Separation Ring
Camera Deployment Mechanism
UHF Antenna Deployment Mechanism
Power
Primary PCDU
Redundant PCDU
Battery Pack
2x Fixed Deployable Solar Panels
2x Body-Mounted Solar Panels
Separation Switch
Remove-Before-Flight Switch
ADCS
ADCS Controller (incl. IMU)
Sensors:
2 x Fine Sun Sensors
2 x Star Trackers
Magnetometer
2 x GPS Antenna + 2 x Receiver
Actuators:
4x Reaction Wheels
3x Magnetorquers
Propulsion
5N Thruster
Hydrazine Propellant Tank
Nitrogen Pressurant Tank
Pressure Regulator
Feed Valves & Lines
Comms/OBDH
2 x S-band Patch Antennas
2 x S-band Transceiver
2 x UHF Monopole Antennas
2 x UHF Transceivers
RF Switch
OBC
Thermal
Aluminised Kapton
Silver-coated FEP
Heat Pipes
Heaters
Harnessing
Main Satellite Harness
Primary Payload
Space Forge CVD Payload
Secondary Payloads
Microgravity Sensor
Cold Cathode Pressure Transducer
Satellite Selfie Camera